

Which Imagined Concepts Can Be Steered in LLMs?

Anonymous ACL submission

Abstract

Steering vectors are widely used to control and interpret LLMs, yet benchmarks show that some concepts steer reliably while others barely respond — a variance that is reported but rarely explained. In this research we investigate which kinds of concepts are steerable in Llama-3.1-8B and Gemma-2-9B. We use the RFM method to extract steering vectors (or directions) for approximately 600 concepts drawn from 24 WordNet supersense categories, and validate which of them successfully steer the model. We then build a regression model explaining steerability from concept category, concreteness, polysemy, and word frequency, as well as the share of the top eigenvector during extraction (λ_1 /trace). The results help distinguish concepts that behave as single directions — “primitive” for the model — from those that are composite or context-dependent, and offer practitioners an extraction-time indication of when single-vector steering can be trusted.

- TODO: the prose paragraph above pre-dates the final results — update: add age-of-acquisition to the predictor list, replace “ λ_1 /trace” with “spectrum concentration (participation ratio)”, and consider adding one sentence each on the depth-window finding and the cross-model replication.
- Headline: 41.5% of the 600 concepts steer on Llama (249/600, V1UV6) and 37.3% on Gemma (224/600) under the v4 instrument, and the psycholinguistic laws replicate across BOTH models with all four predictors significant: frequency +, concreteness –, age-of-acquisition –, polysemy –. Spectrum concentration (PR) predicts steerability within both models, at model-specific depths. Frequent, early-acquired, monosemous, abstract concepts steer best.

- Candidate sentence (failure mode): when steering fails, it fails informatively — delivery of the category/frame blend, a diffuse discriminative spectrum, or injection outside the concept’s depth window; not silence. (1 sentence)
- Candidate sentence (instrument): measured steerability is instrument-relative — one changed frame sentence doubles the rate on both models — but the laws survive instrument and model change. (1 sentence)
- Candidate sentence (depth): concepts are injectable in depth windows ordered by abstractness; one well-chosen layer nearly matches all-layer steering. (1 sentence)

1 Introduction (1.0p)

- Hook: steering/RePE papers throw heterogeneous concept lists at models; some concepts steer at 90%, others at chance. The variance is treated as noise; we argue it is signal. (1 para)
- The implicit assumption of top-1 eigenvector / PCA-1 / difference-of-means steering: the concept is representable as one direction, injectable with one coefficient, at whatever layer(s) the method uses. (half para)
- Existing evidence that not all features are one-dimensional (Engels et al.) — but no one has asked which concepts behave as single directions, or whether this aligns with human semantic structure. (half para)
- Our question: is steerability systematic and predictable from the linguistic properties of the word — and what does a steering failure actually consist of? (half para)
- Contributions (short lead-in + three numbered items, 2–3 sentences each): (1 para)

A Same protocol, opposite outcomes (steered layer set, rising coefficient)

'frog' (steered)

c=0.5: I'd love to chat about the latest space discoveries! Just returned from a galactic adventure, eager to ...

c=0.6: "Ribbit! Oh, I'd love to chat about the juicy bugs I caught in the pond today! Squirm, squirm!"

c=0.7: "Splash-splash! Oh, I'm so glad to be a frog on a lily pad on a lazy summer day by the pond!"

c=0.8: "Squirm! Ribbit! Oh, the sun is shining bright on my lily pad today! Ribbit!" (Hee-hee! Hoppy to chat with ...)

'fairway' (not steered)

c=0.5: The latest advancements in space exploration and the potential for human settlements on Mars. Exciting ...

c=0.6: Lovely summer day! Perfect weather for a stroll along the beach, feeling the warm sun on my skin, so ...

c=0.7: Lovely summer day! Perfect weather for a golf game on the lush green course with the warm sunshine on my face.

c=0.8: "Welcome to the 26th day of July, 2024! Perfect day for a picnic on the picturesque green golf course!"

B Which concepts steer varies systematically by category (600 WordNet concepts, Llama-3.1-8B)

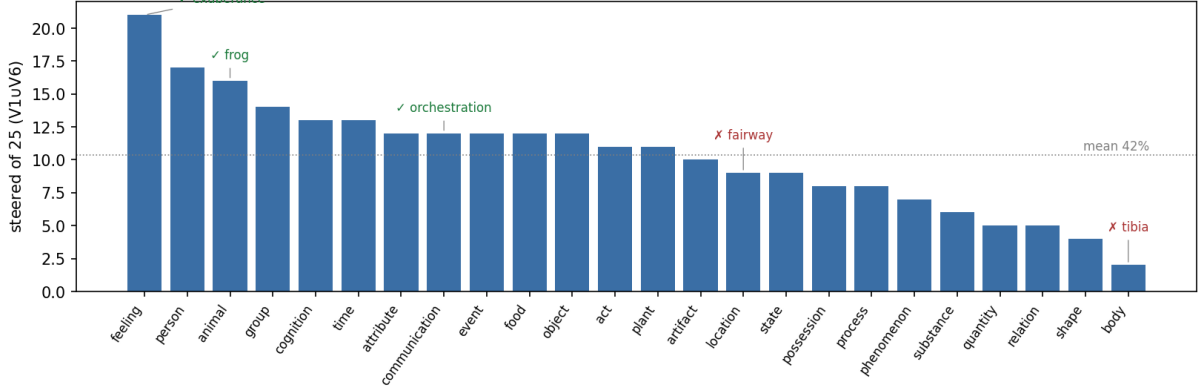


Figure 1: (A) Same protocol, opposite outcomes: dose-response for a steerable concept (frog) vs. an unsteerable one (fairway); generations from the neutral evaluation prompt (V6), all-layer steering, Llama-3.1-8B, v4 frame. (B) Concepts steered per WordNet supersense (of 25, V1UV6): steerability varies systematically by category.

- C1: a systematic steerability study — ~600 lexicalized concepts across 24 WordNet supersenses, annotated with psycholinguistic norms, steered under a controlled protocol on two models — together with a 7-frame instrument study (Table 1) showing that the extraction contrast itself sets measured steerability (2× rate swing at fixed method and model); per-concept results and code released.
- C2: a regression explaining steerability from frequency, concreteness, age-of-acquisition, and polysemy (all significant, signs replicating across Llama and Gemma), with extraction-internal spectral concentration (participation ratio / λ_1 -share) as a zero-cost predictor that beats the lexical block.
- C3: a mechanistic account of what failure looks like — parent-dominance of extracted directions, coherence $\neq$ iden-

tity, injection outside the concept’s depth window — probed causally (within-category subtraction; single-layer depth sweep).

2 Background and Related Work (1.0p)

- Linear representation hypothesis and steering. Park et al. (2024); RepE (Zou et al., 2023); CAA (Panickssery et al., 2024); ActAdd (Turner et al., 2023); ITI (Li et al., 2023); the RFM benchmark we build on (Beaglehole et al., 2025): kernel ridge + AGOP, top eigenvector as steering vector. Ours: we adopt RFM unchanged and ask which concepts it works for. Second relation: steering rates are instrument-relative — one changed contrast sentence doubles the rate (19.8%→41.5%) at fixed method, model, and judge; method comparisons with hand-tuned per-class prompts inherit this confound. (1 para)
- Multi-dimensional features. Engels

et al. (2025): not all features are one-dimensional. Ours: we test the premise on a stratified inventory and tie it to word properties: spectrum concentration (PR / λ_1 -share) predicts steering in both models — “effectively one-dimensional” is measurable and graded, and polysemy acts on steerability largely through it. (half para)

- Hierarchical concept geometry. Park et al. (2025): ~ 900 WordNet concepts, Llama-3 and Gemma; hierarchically related concepts occupy orthogonal subspaces — the representational-geometry twin of this study. Ours: they characterize how hierarchy is encoded; we test what it predicts about intervention. Our parent-dominance finding is the steering-side counterpart of their parent/child decomposition; and our R4 probe reads their orthogonality causally — the child-specific residual exists but is rarely independently injectable (9 gained / 14 lost of 150). (half para)
- Predicting steerability. Directional agreement and geometric predictors of steering (un)reliability (Braun et al., 2025, 2026) — the closest neighbors to our R3; AxBench (Wu et al., 2025) for large-scale evaluation. Ours: we predict from the concept’s linguistic structure; the lexical laws (frequency +, concreteness –, AoA –, polysemy –) replicate across two models and two instruments — portable predictors engineering signals cannot supply. We add a caution: geometric predictors are only interpretable relative to the contrast design (λ_1 -share flipped null $\rightarrow$ significant with the frame). AxBench positioning: they sample concepts from the model (SAE-latent labels; no norms, no stratification) and vary methods; we sample from the lexicon and fix the method — only the latter can ask which kinds of concepts steer. Their trained ReFT-r1 bounds our claims: “unsteerable” means extraction-based rank-1 steering fails. TODO cite: LAP / logit-lens accessibility (Billa 2026); “When is your LLM steerable?” (bib stubs). (1 para)
- Causal identification of concept vectors. Function vectors (Todd et al., 2024); relation decoding (Hernandez et al., 2024);

ROME (Meng et al., 2022). Ours: we identify by contrast design, probe by intervention (R4). The depth sweep converges with causal tracing: entity content is injectable in a sharp mid-stack window (depths 14–17 of 32), where tracing localizes subject enrichment — but read-side statistics cannot substitute for write-side tests. (half para)

- Frequency and linear representations. Co-occurrence thresholds; long-tail knowledge (Kandpal et al., 2023). Ours: frequency is a robust positive predictor in both models — against our pilot expectation, with co-occurrence-threshold accounts — and thins when spectrum concentration enters: exposure appears to work partly through representational consolidation. (half para)
- LLM–human alignment and psycholinguistic norms. Shani et al. (2025) compare LLM and human concepts on input-side embeddings. Ours: mid-stream and causal — and human structure does predict what can be steered: early-acquired, frequent words steer best (AoA $\beta = -0.33 / -0.49$ across models), over and above frequency. (half para)
- Methodological caution on cosine. You (2025): cosine misleads in anisotropic spaces. Ours: geometry claims rest on differential cosines under identical measurement, with the between-class value (0.35) as the anisotropy control. (half para)
- Gap statement: no prior work connects concept-level semantic structure to steering outcome or explains what failure consists of. Our answer: (i) frequency, AoA, polysemy, and concreteness jointly predict steerability, replicating across Llama and Gemma; (ii) failure is usually informative — category/frame delivery, a diffuse spectrum, or injection outside the depth window — not silence. (half para)

3 Defining and Sampling Concepts (1.0p)

- Operational definition. Sidestep the philosophical debate (Murphy 2002; Margolis & Laurence 1999): a lexicalized concept = a meaning a speech community assigned a word to, operationalized as a WordNet

228	synset; each word mapped to its dominant	278
229	(first) noun sense. (half para)	279
230	• Inventory. All single-word, noun-dominant	280
231	lemmas from WordNet; category = super-	281
232	sense (26 lexicographer files; Miller 1990;	282
233	Fellbaum 1998; “supersense”: Ciaramita	283
234	& Johnson 2003). Exclusions: noun.Tops;	284
235	POS-ambiguous lemmas; small cells sam-	285
236	pled exhaustively and flagged. (half para)	286
237	• Sample (≈ 600 , flat). 25/supersense across	287
238	~ 24 usable supersenses; within each su-	288
239	persense, stratify over frequency so cate-	289
240	gory and frequency are not confounded.	290
241	Six supersenses are pre-specified (before ex-	291
242	traction) as the featured cells for within-	
243	category analyses: location, person, feeling	292
244	(pilot continuity) and artifact, cognition,	293
245	food (concreteness extremes). At $n=25$	294
246	every cell is adequately powered for class	295
247	means and absorbs attrition (prompted-	296
248	ceiling/judge exclusions). (1 para)	297
249	• Predictors (one table): Zipf frequency	
250	(Speer et al., 2018); concreteness (Brys-	298
251	baert et al., 2014); age of acquisition (Ku-	299
252	perman et al., 2012); polysemy (#Word-	300
253	Net senses); supersense; log category size	301
254	(#noun synsets in the lexicographer file,	302
255	covariate). Tested and null: token count,	303
256	verb-sense count. (half para + table)	304
257	• Contrast design. Positives wrap a state-	305
258	ment in a universal frame; the frame must	306
259	be felicitous across all supersenses, else	307
260	frame felicity confounds category; nega-	308
261	tives are the bare statements (benchmark-	309
262	comparable). Rationale for care here: a	310
263	contrast in which frame, category, and in-	311
264	stance always co-occur cannot statistically	312
265	separate them (see Pilot) — whatever the	313
266	frame adds constantly ends up inside every	
267	direction. A second negative design — same	314
268	frame with a different same-supersense con-	315
269	cept, which cancels frame and category par-	316
270	ent — is used as the causal probe (R4), not	317
271	the main contrast. (1 para)	318
272	• [NEW jul10] Main-run frame: v4 = “Imag-	319
273	ine/assume: {c}.” (constant prefix, con-	320
274	cept unquoted; negatives bare). Selected	321
275	from a 7-frame grid on the pilot (Table 1,	322
276	Appendix F): 77/90 V1UV6 vs. 61 (orig	323
277	class-specific) and 59 (v2b dual frame) —	324
	the only frame at ceiling on moods and	325
	far ahead on places. Ablations attribute	326
	the gain to assume (bare-noun reading	
	$\approx$ role-assignment, a class-neutral persona	
	prime) and show quoting the concept costs	
	~ 7 points (mention-mode suppresses enact-	
	ment). Trade-off, quantified in Appendix F:	
	a constant prefix re-saturates extraction R^2	
	(the prefix is a free class separator), so the	
	dual-frame diagnostics (informative val- r)	
	are given up in exchange for steering power;	
	the frame-leak caveat transfers to the imag-	
	ine/assume register (acknowledged in Lim-	
	itations).	
	• Pilot/confirmatory split. Pilot = 90 con-	
	cepts (30 each: places, moods, personal-	
	ities) from the released benchmark lists,	
	used for all exploration; the WordNet sam-	
	ple is confirmatory, untouched during ex-	
	ploration. (half para)	
	4 Method (1.0p)	
	• Direction extraction (RFM) — and why	
	RFM. Per layer: kernel ridge regression on	
	concept-vs-negative prompts; AGOP $M =$	
	$\mathbb{E}[\nabla f \nabla f^\top]$; steering vector = top eigen-	
	vector of M . Chosen among RepE meth-	
	ods because (a) it is the method of the	
	benchmark we build on (comparability); (b)	
	unlike difference-of-means or PCA-1 it re-	
	turns the full discriminative spectrum with	
	eigenvalues — the spectral metrics below	
	come for free, and the method is self-	
	diagnosing (the pilot’s confound was discov-	
	ered through them); (c) its steering perfor-	
	mance matches or beats the simpler extrac-	
	tors on the released benchmark. (1 para)	
	• Spectral metrics. $\lambda_1/\text{tr}(M)$, λ_1/λ_2 , and	
	the participation ratio $\text{tr}(M)^2/\ M\ _F^2$; per-	
	concept score read at a fixed relative depth	
	(per-layer profile retained; the informa-	
	tive depth is model-specific — 40% on	
	Llama, 60% on Gemma). These measure	
	the shape of the discriminative spectrum,	
	not concept identity (see Pilot); they are	
	near-deterministic functions of one another	
	($ r \geq .94$), so they enter analyses one at a	
	time. (half para)	
	• Fixed hyperparameters — a methodolog-	
	ical requirement. Kernel bandwidth in-	

frame	class	V1	V4	V6	V1UV6	med R^2
orig (class-specific)	pers.	24	4	17	26	.996
	moods	30	27	27	30	.996
	places	2	14	4	5	.998
v2 “fascinated / preoccupied”, quoted	pers.	24	3	9	24	.804
	moods	22	16	24	26	.740
	places	10	4	3	11	.807
v2b “fascinated / expert on”, quoted	pers.	22	–	13	23	.715
	moods	17	–	14	19	.785
	places	14	–	13	17	.531
v3 “thinking about / on your mind”, quoted	pers.	18	0	8	20	–
	moods	16	13	9	17	–
	places	7	9	5	8	–
v4 “Imagine/assume: {c}.”	pers.	23	9	9	23	.997
	moods	29	28	27	30	.995
	places	16	17	22	24	.992
v4b “Imagine: ‘{c}.’” (quoted)	pers.	20	2	10	23	.997
	moods	19	20	14	22	.986
	places	10	13	12	14	.990
v4c “Imagine: {c}.”	pers.	23	4	10	25	.997
	moods	28	27	24	28	.993
	places	8	12	12	13	.994
v4d “Imagine/consider: {c}.”	pers.	21	3	5	21	.996
	moods	21	23	17	22	.985
	places	10	10	13	13	.991

Table 1: Extraction-frame grid (pilot $N=90$: 30 personalities / 30 moods / 30 places; Llama-3.1-8B, all-layer $K=1$, default coefficients, gpt-oss judge). Cells = concepts steered out of 30 under each evaluation version; V1UV6 = union outcome; med R^2 = class median of per-concept median validation R^2 across layers. Constant-prefix frames (orig, v4 family) saturate R^2 by construction (the prefix is a free class separator); dual frames (v2, v2b) make R^2 concept-informative; v3 direction files were not retained, so its R^2 cells are unavailable. V1UV6 totals: orig 61, v2 61, v2b 59, v3 45, v4 77, v4b 59, v4c 66, v4d 56.

flates λ_1 /trace mechanically (large bandwidth $\Rightarrow$ near-linear fit $\Rightarrow$ rank-1 AGOP); grid search over near-tied validation correlations injects artifact variance into the dependent measure. All extractions at bw=1, norm=True, reg= 10^{-3} ; vector and metrics read at a fixed (converged) iteration. (half para)

- Two-config steering protocol. Each concept steered under (i) all layers, benchmark defaults (unit vectors, released coefficient grid — comparability); (ii) one mid layer at $\approx 40\%$ depth (vector scaled by the concept’s per-layer projection gap, cf. CAA / ITI; effective single-site dose $\approx 1 \times$ local residual-stream norm). Δ (all – single) measures how distributed the concept’s steerable content is — with the caveat, added after the depth sweep, that Δ partly measures window placement: the confirmatory single-site arm should sit at depth 15, inside the entity window. Two evaluation prompts (category-inviting V1; neutral V6; listed verbatim); greedy decoding. (1 para)
- Outcome. Primary: steered under V1 or

V6 at any coefficient (binary; the union de-biases the evaluation-relativity of either prompt alone, V1–V6 $\phi=0.59$). Secondary: granularity reached (category / region / instance) — the binary criterion flattens near-misses (pilot: Edinburgh $\rightarrow$ “Scottish Highlands” scores 0); a graded rubric (V8) re-judges the same generations as a robustness check (Appendix B). TODO: the category/region/instance granularity coding was never systematically performed on the 600-run — V8 partially covers it; either promote V8 to fill this role explicitly or cut the granularity promise. TODO: the evaluation prompts and the v4 frame are described but never listed verbatim — add a prompts appendix (frame, V1, V6, judge templates). (1 para)

- Judge. gpt-oss:20b (local, on the experiment GPU); symmetric judge question across supersenses. Validated against Claude Sonnet 5 on a 360-item set: 87–91% agreement, $\kappa = .72-.78$; a stricter candidate (Haiku 4.5) was rejected for one-directional divergence from the two-

377	judge consensus. Confirmatory-study ad-	the category. (1 para + figure par-	426
378	ditions: re-validation on ~ 200 WordNet-	ent_share_by_depth.png)	427
379	distribution items + human spot-check of		
380	disagreements (Appendix E). (half para)		
381	• Controls. Prompted ceiling (can the	• Finding 2: coherence $\neq$ identity. Places	428
382	model express the concept when explicitly	show the highest band λ_1 /trace yet fail	429
383	asked?) — steerability conditional on ceil-	single-layer steering; moods (lower) suc-	430
384	ing; pipeline sanity-checked on known-good	ceed. Spectral metrics certify that a co-	431
385	concepts per model. TODO: the prompted-	herent (context-invariant, rank-1) direction	432
386	ceiling control is promised here but has not	was extracted — not that it is the judged	433
387	been run at $n=600$ — either run it (one	content. Consequence: their regression co-	434
388	cheap generation pass, no steering) or cut	efficients are expected to be conditional on	435
389	the promise. It matters most for the bottom	contrast type; tested under both contrasts.	436
390	cells: is noun.body unsteerable or merely	(1 para)	437
391	unpromptable? (half para)		
392	• Models. Full protocol on Llama-3.1-8B-	• Finding 3: attribute vs. instance. Attribute	438
393	Instruct; headline analyses replicated on	concepts (moods) steer from one mid layer;	439
394	Gemma-2-9B-it. (half para)	instance identities steer only under all-layer	440
		injection, reaching region granularity —	441
395	5 Pilot Diagnostics and Protocol	consistent with multi-stage assembly (per-	442
396	Selection (1.0p)	layer residue accumulation + late lexical	443
		tipping + autoregressive bootstrap once the	444
397	• Hyper-parameter selection (summary).	name is emitted). Motivates the two-config	445
398	Validation r saturates (~ 0.97 – 0.99) across	protocol and Δ . (1 para)	446
399	bandwidths, statement counts, and it-		
400	erations $\Rightarrow$ selection by validation is	• Protocol justifications (one line each; evi-	447
401	third-decimal noise; large bandwidth forces	dence in Appendices C–D): single mid layer	448
402	rank-1 AGOP by construction; converged	in the mid-stack behavioral window — the	449
403	readout cuts metric instability 3–4 $\times$. De-	depth sweep (Discussion) later located it	450
404	cision rule: fixed modal setting, converged	precisely at depths 14–17 for entity con-	451
405	readout. Details in Appendix A. (NB the	cepts, with the original 40% (depth-13) site	452
406	saturation itself is frame-structural: any	just below its onset; 60% yields token-bias	453
407	constant-prefix contrast hands the classifier	then repetition collapse. No top- k arms —	454
408	a free separator; dual frames de-saturate	the top-3 subspace under the released con-	455
409	val- r — Appendix F.) (half para)	trast is category-shared. Coefficient $\approx 1\times$	456
410	• Finding 1: parent-dominance. Different	local residual norm via per-layer magn scal-	457
411	same-category concepts’ extracted top-1 di-	ing. Granularity as secondary outcome —	458
412	rections are nearly identical (places: co-	the binary judge flattens near-misses. (1	459
413	sine $0.97 \rightarrow 0.84 \rightarrow 0.72$ across depth; all	para)	460
414	top-3 components ≥ 0.90 shared). Single-		
415	layer dose-response saturates at the cat-	6 Results (1.5p)	461
416	egory prototype (every city $\rightarrow$ “quiet		
417	beach at sunset”) with region-level near-	• R1: Steerability varies systematically —	462
418	misses (Edinburgh $\rightarrow$ Dublin pub); giving	249/600 (41.5%) steer (Llama, all-layer, v4	463
419	secondary components injection mass adds	frame, judge = gpt-oss; V1 191, V6 195,	464
420	a generic roleplay axis, not instance con-	union V1 $\cup$ V6; Figure 3). Per-supersense	465
421	tent. Three legs: geometry, dose-response,	spread 2–21 of 25: top noun.feeling	466
422	component ablation. Interpretation: a	(21), noun.person (17), noun.animal (16);	467
423	contrast in which frame, parent, and in-	bottom noun.body (2), noun.shape (4),	468
424	stance always co-occur is statistically one	noun.quantity/noun.relation (5). (1 para +	469
425	axis — what steering delivers is mostly	figure)	470
		• R1 instrument comparison (v2b arm): the	471
		frame moves the rate, much less the order-	472
		ing. Under the dual v2b frame the same	473

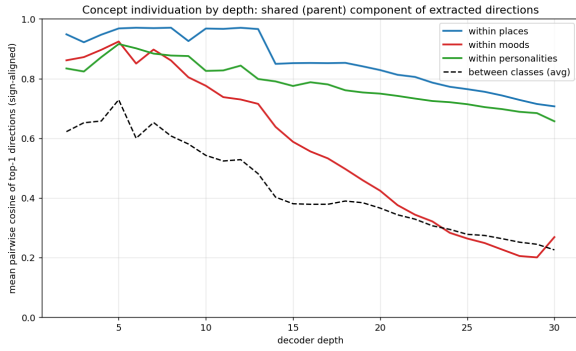


Figure 2: Pilot Finding 1 (parent-dominance). Cosine similarity of same-category concepts’ extracted top-1 directions across depth: the released contrast yields largely shared (category) directions for places, while moods individuate.

protocol yields 119/600 (19.8%; spread 2–9); switching to v4 improves 22 of 24 supersenses while the ranking stays broadly stable (feeling top, body bottom under both; noun.body immovable at 2/25). V1–V6 agreement $\phi = 0.59$ — measured steerability is partly evaluation-relative, motivating the union outcome. (half para)

- R2: Psycholinguistic predictors (Table 2): all four lexical predictors significant, same signs in both models. Llama: conc -0.20^* , zipf $+0.28^*$, AoA -0.33^{**} , polysemy -0.30^{**} ; Gemma: conc -0.20^* , zipf $+0.23^\dagger$, AoA -0.49^{**} , polysemy -0.35^{**} . The pilot expectation (concreteness +, polysemy carrying the signal, frequency null) is rejected: frequent-but-abstract concepts steer best. AoA and polysemy were null under v2b and surface once the instrument improves (v2b: zipf $+0.40^*$, conc -0.28^* ; Appendix B). Honest-reporting nuance (Figure 4): polysemy and concreteness do not separate univariately (MW $p > .6$; effects conditional on correlated covariates); zipf, AoA, PR separate cleanly in both models ($p < .005$). Collinearity among lexical predictors is benign (conc–zipf $r = .09$; VIF < 2.7). Log category size is a positive predictor in Llama ($+0.31^{**}$ with the lexical block; $+0.21^*$ surviving PR) but null in Gemma (-0.01) — reported as a non-replicating candidate, not among the laws. (1 para + table)
- R3: Spectral metrics: every spectral met-

ric carries signal, and each alone beats the whole lexical block. Llama, one-at-a-time: $\lambda_1/\text{trace} +0.55^{**}$ (AIC 778.5), $\lambda_1/\lambda_2 +0.57^{**}$ (775.3), PR -0.49^{**} (786.5) vs. lexical block 799.0, null 816.4; best model lex+PR (763.6, PR -0.62^{**}). Concentrated discriminative spectra steer; polysemy is largely absorbed by PR in the joint model — polysemous words appear to fail via diffuse spectra. Instrument caution (one of our sharpest methodological findings): under the dual v2b frame λ_1/trace was null — the register axis parked in within-positive variance can capture the top eigenvector for weak concepts (Appendix F), corrupting spectral readouts. Spectral diagnostics are only interpretable relative to the contrast design. Spectral metrics are near-collinear ($r \leq -.94$): one-at-a-time only. (1 para)

- Category is partly mediated by measurable word properties. Supersense dummies beat the null (AIC 784.1 vs. 816.4) but lose to lexical+PR (763.6), and adding them to the best model does not pay: the category signal visible in R1 is substantially carried by the continuous predictors. (Under v2b the dummies did not even beat the null; the wider v4 spread carries real categorical signal.) (half para)
- R4: The rescue probe (causal). Design: for failing concepts, cancel the frame/parent component — either by re-extracting with within-category contrasts (“loves Edinburgh” vs. “loves {other city}”; not yet run) or by subtracting the leave-one-out supersense mean from the saved direction (v_{spec} ; executed, $n=150$). Outcome: no net rescue (25 vs. 30 of 150 steer; McNemar $p \approx .4$) — shared-component dominance is not the prevailing failure mode, though the subtraction doubles as a frame-leak control. Details moved to Appendix Z3 [bold note: full v_{spec} material relocated there; inclusion in the final submission undecided]. (1 para)
- R5: Distributedness. $\Delta(\text{all-layer} - \text{single-layer})$: single-layer depth-13 (magn dose) 90/600 (15.0%) vs. all-layer 249 (41.5%), ratio $2.8\times$ (v2b: $3.5\times$); 11 concepts steer only single-layer (over-steer rescues; ab-

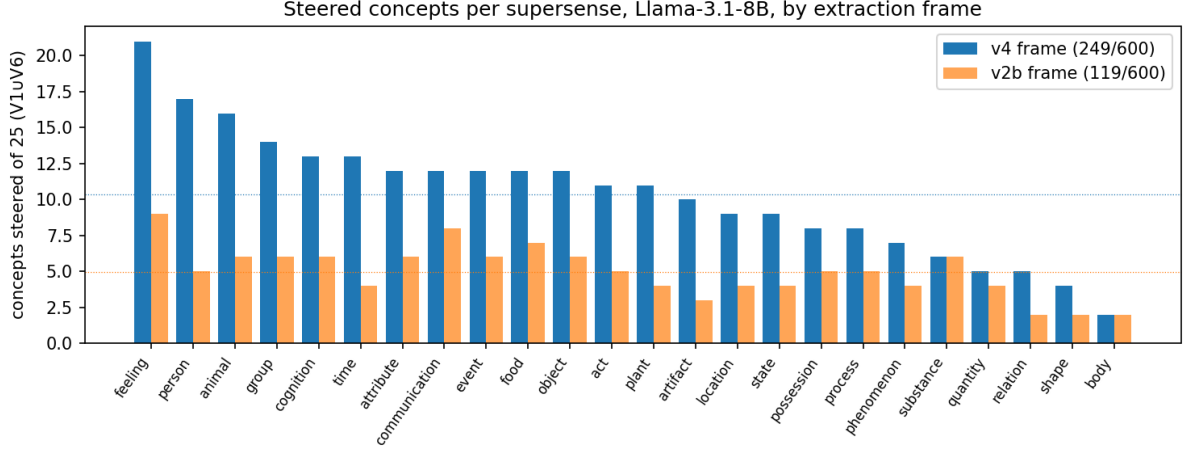


Figure 3: [NEW jul10] R1: steered concepts per supersense (of 25; V1uV6; Llama-3.1-8B, all-layer config), under the main v4 instrument and the v2b instrument-comparison arm. The frame moves the rate ($\sim 2\times$), far less the ordering (feeling top, body bottom under both).

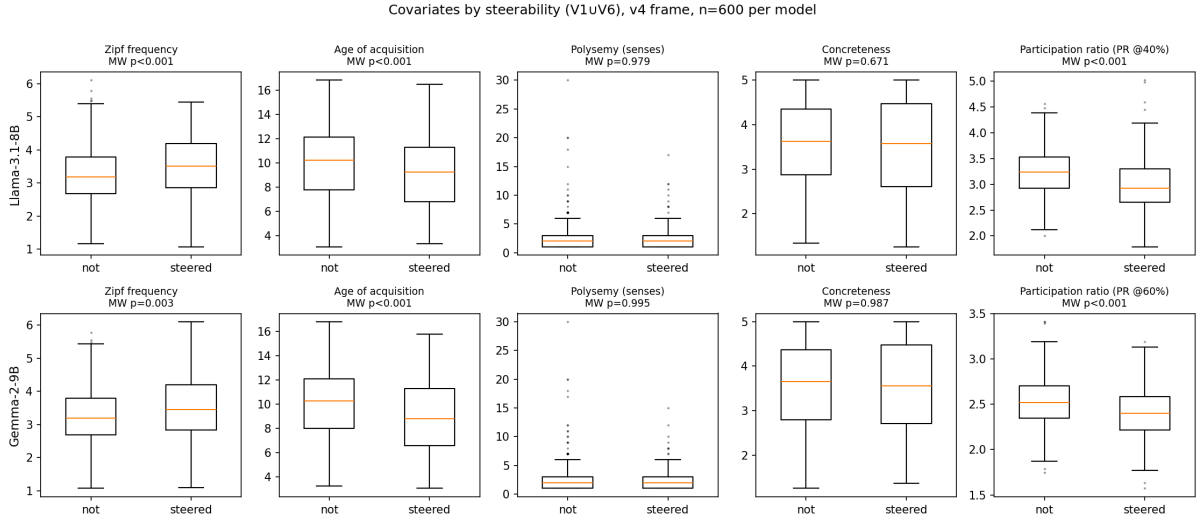


Figure 4: [NEW jul10] R2/R3: covariates by steering outcome (v4 runs, $n=600$ per model; PR at each model’s informative depth). Zipf, AoA, and PR separate univariately in both models; polysemy and concreteness do not (their regression effects are conditional on correlated covariates).

stract/attribute skew). Caveat from the depth sweep: depth 13 sits below the entity window, so Δ partly measures window placement — interpret jointly with the sweep (Discussion). (half para)

- R6: Generality (Gemma-2-9B replication): rates and laws replicate; item convergence is itself instrument-dependent. Rates: 224/600 (37.3%) vs. Llama 249 (41.5%); the v4 gain replicates ($\sim 2\times$ both models’ v2b rates). Laws: all four lexical signs replicate (AoA -0.49^{**} strongest); PR replicates at Gemma’s informative depth (60–70%: -0.42^{**} ; null at Llama’s 40%

convention) — same law, model-specific depth. Items: per-concept $\phi = 0.43$ (154 shared) and supersense-profile $r = 0.44$, both up from v2b (0.33 / 0.12 n.s.): part of the earlier item-divergence was instrument noise, though which concepts steer remains substantially model-relative (Gemma tops: event/phenomenon/group; body bottom in both). Supporting pilot evidence: the λ_1 /trace class ordering inverts across models (Llama places $>$ moods; Gemma the reverse) — spectral values are model-geometry-laden; compare within-model only. Gemma dose grid [14,17,20,23]

	Llama-3.1-8B		Gemma-2-9B	
	lexical	+PR	lexical	+PR ^{60%}
concreteness	-0.20*	-0.43**	-0.20*	-0.18†
Zipf frequency	+0.28*	+0.23	+0.23†	+0.20
AoA	-0.33**	-0.28*	-0.49**	-0.47**
polysemy	-0.30**	-0.15	-0.35**	-0.29*
n tokens	+0.04	-0.05	+0.06	+0.02
participation ratio		-0.62**		-0.39**
AIC	799.0	763.6	767.3	750.9

Table 2: Main regression (v4 instrument). Logistic regression of steerability ($V1 \cup V6$) on z-scored predictors, $n=600$ per model, all-layer config. PR read at each model’s informative depth (Llama 40%, Gemma 60% — Gemma’s spectral signal is null at the 40% convention and switches on at 60–70%). Reference AICs — Llama: null 816.4, supersense-only 784.1, one-spectral-alone 775–787; Gemma: null 794.8. Spectral collinearity is extreme under v4 ($PR-\lambda_1/\text{trace } r \leq -.94$ both models): spectral metrics are reported one-at-a-time only. † $p<.1$, * $p<.05$, ** $p<.01$.

set by dose-response audit (Appendix D).
(1 para)

7 Discussion (1.0p)

- What makes a concept steerable: attributes (context-general colorings) admit a single injectable direction; instance identities require assembled, multi-stage content — steering failure is usually delivery of the parent category, not silence. (1 para)
- What the spectral metrics are: an intervention-match score, with an instrument caveat. Spectrum concentration (low PR / high λ_1 -share) measures how well the concept’s discriminative signature fits the one move single-vector steering can make (one constant vector, all contexts); it predicts steering in both models. The mechanism behind R2’s reversal: abstract (not concrete) concepts steer better — for concrete instance-like concepts the dominant direction is largely the shared category axis, so its coherence does not deliver the instance (parent-dominance). And the caveat: spectral diagnostics of extracted directions are only interpretable relative to the contrast design that produced them — under the dual v2b frame λ_1/trace read as null because a frame-register axis could capture the top eigenvalue slot (Appendix F);

under the constant v4 prefix it predicts steering with the expected sign. (1 para)

Measured steerability is instrument-dependent — and so is cross-model generality. Changing one frame sentence doubles the steerable share in both models (19.8→41.5% Llama; 18.8→37.3% Gemma) and raises cross-model item agreement (ϕ 0.33→0.43) and category-profile agreement (r 0.12→0.44). Absolute rates from any single protocol should not be read as properties of the model alone; but the laws (frequency +, concreteness –, AoA –, polysemy –, spectrum concentration –) survive instrument change and model change, which is the paper’s strongest generality claim. (half para)

- Depth structure: concepts live in depth windows, ordered by abstractness. Attributes are injectable from ~30% depth onward with a broad plateau; entity identities switch on sharply at ~45% (window 14–17 of 32) and fade late; personas need multi-site injection. One well-chosen layer nearly matches all-layer steering (71 vs. 77 of 90) — “all layers” mostly buys window coverage. Read-side extraction statistics do not locate the window (they miss the late decline entirely): injectability is a write property. (half para)
- Status of extracted vectors: the residual stream has no privileged basis, so extracted directions are experiment-relative; within-model relational claims are valid, coordinate-level and raw cross-model comparisons are not. Anchored spaces (embedding/unembedding, MLP neurons) are where mechanism pins a frame. (half para)
- Two routes to a vector’s identity: experiment design (contrasts that vary exactly one thing) vs. causal mediation (function vectors, ROME). Pipeline: propose by contrast, accept by causal test — R4 instantiates it. (half para)
- Practical implications for RePE: report spectral concentration and contrast design alongside steering benchmarks; stratified concept inventories; granularity-graded judging over binary. (half para)

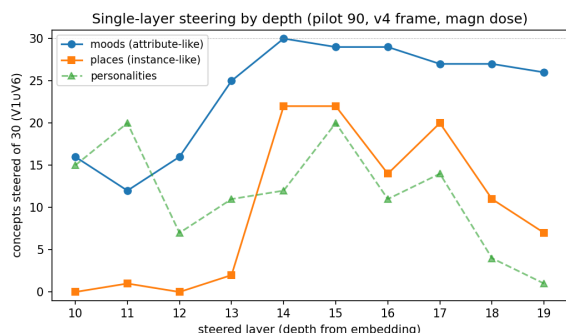


Figure 5: [NEW jul10] Retention by steered depth (pilot 90, v4 directions, single-layer magn dose). Attributes (moods) steer shallow and everywhere; entity identities (places) switch on sharply at depth 14 (of 32), window 14–17; personas never find a single-layer home.

The depth axis: first-steerable-depth sweep (0.5p)

- Design. Pilot 90 (moods/places/personalities) × every single-layer depth 10–19, magn dose, v4 directions, V1UV6 (Figure 5). (half para)
- Result 1: entity concepts have a sharp onset at depth 14 (of 32). Places: 0–2/30 at depths 10–13, then 22/30 at depth 14, window 14–17, fading by 19. This dissolves the earlier “places don’t steer single-layer” conclusion: the old two-site protocol (13/19) straddled the window. (half para)
- Result 2: attribute concepts steer shallow and everywhere. Moods: 16/30 already at depth 10, plateau 25–30 from depth 13 through 19 — attributes become injectable ~ 4 layers before instances (the abstraction-depth hypothesis, confirmed ordinally). Personalities never find a single-layer home (peak 20, collapse to 1 by depth 19) — persona content appears genuinely multi-site. (half para)
- Result 3: one well-chosen layer $\approx$ all layers (71/90 at depth 15 vs. 77/90 all-layer): the all-layer advantage is largely not knowing the window, not distributed injection per se (except personalities). Per-layer read-side metrics rise into the window but entirely miss the late decline ($|r| \approx .5\text{--}.6$): first-steerable-depth is a write property. (half para)
- Future work: same sweep on the

six featured WordNet cells (150 concepts) with taxonomic depth as ground truth (hypernyms shallower than hyponyms?); independence from the spectral axis assessed within supersense; compositional vignette — is direction(“strict father”) $\approx \alpha$ direction(“strict”) + β direction(“father”) (half para)

8 Limitations (0.5p; uncounted in ACL format)

- Single extraction method (RFM); λ_1 /trace is AGOP-specific — PCA explained-variance as cross-check (appendix if time). (half para)
- Steering conclusions are relative to the additive intervention class; clamp/projection interventions may behave differently (targets saved, untested). (half para)
- LLM judge (validated but imperfect); English only; dominant-sense assumption; norms are human ratings. (half para)
- Correlational core with one causal probe (R4); extraction-design effects and model limits not fully separable outside that design. (half para)
- Depth-sweep evidence is pilot-only ($n=90$, 3 concept classes, one model); the WordNet-wide sweep and the Gemma sweep are future work. Measured rates are instrument-relative (quantified, but a reader should not quote “41.5%” as a model constant). (half para)
- The v4 frame’s constant prefix puts an imagine/assume register component into every direction; the frame-leak control (v_{spec} subtraction, Appendix Z3) was run on the v2b instrument only and not repeated for v4 — some v4 successes may be partly frame-carried. (half para)

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		also a poor steerability predictor. Boxplots:	849
		box_rsqr.png.	850

+ λ_1 /trace at bandwidth $L=1$ discriminates across concepts; $L=10$ collapses it toward 1 (rank-1 AGOP by construction). `box_l1trace.png`, `box_l1l2.png`.

+ The released λ_1 /trace (read at the best- r iteration) is unstable — it samples a value climbing $0.0002 \rightarrow 0.17 \rightarrow 0.46$ across iterations at a noise-chosen point; reading at convergence cuts the statement-count swing 3–4 $\times$. Both vector and metrics read at a fixed (converged) iteration.

+ Saved per-layer extraction stats: top-3 eigenvalues (evals), natural concept magnitude ($\text{magn} = \text{raw projection gap}$), clamp targets (tpos , tneg), mean residual norm (hnorm) — in the `rfmstats` pickles.

? Per-concept “best layer” by `argmax` is unstable: $\text{val-}r$ best is noise-driven; λ_1 /trace and λ_1/λ_2 are bimodal with a degenerate spike at the first block (exclude early blocks). `besthist_*.png`.

? Hyper-parameter sweep factors (probe set of 24 concepts, one per supersense): bandwidth $\times$ normalization $\times$ contrastive construction (now incl. within-category negatives) $\times$ iteration readout; criteria: stability, discrimination, semantic signal (Spearman vs. concreteness/polysemy), steering non-degradation. Artifacts: `hp-sweep_phase2_*`.

B Appendix B: Predicting steerability (pilot 3 classes + 600-concept main run)

+ Best single predictor of steering success is λ_1 /trace at a mid layer (depth 9, grouped CV-AUC ≈ 0.86); λ_1/λ_2 similar; $\text{val-}r$ erratic (saturated). Logistic regressions in `nb_regxp1.ipynb`; `L9_metrics_by_steerability.png`.

– Multivariate regression on all layers’ metrics is uninformative — severe multicollinearity (the residual stream is cumulative). Univariate per-layer CV-AUC is the clean read.

+ Sign flips with depth: λ_1 /trace predicts steerability positively early (depth 9) but

negatively late (depth 29) — late-layer concentration is lexical collapse, not concept concentration. `l1l2_depth29_dist.png`.

? Concept class is a confounder/mediator: most steerability variance is between-class; within-class signal is weak. Do not “control for class” if class is on the causal path (concreteness $\rightarrow$ signature coherence $\rightarrow$ steerability).

? The inversion: places have the highest band-averaged λ_1 /trace yet fail single-layer steering, while moods (lowest) succeed — the metric certifies coherence of the extracted blend, not its identity. Predictor validity is contrast-dependent (main text, Pilot Finding 2).

? Participation ratio $\approx 2\text{--}3$ for all concepts at all layers — the discriminative subspace at one layer is effectively rank 2–3, bounding what top- k can recover. `PR_depth9_hist.png`.

+ Error analysis at depth 9: misclassifications symmetric, in a narrow overlap band; characterizes the ~ 0.86 ceiling.

+ v2b-instrument regressions (Table 3; superseded as main numbers by the v4 runs). Under the dual v2b frame: $\text{zipf} + 0.40^*$, $\text{conc} - 0.28^*$; AoA/polysemy null; PR the only working spectral metric (-0.27^* alone; λ_1 /trace null and sign-flipped — later diagnosed as frame-axis eigenvector capture, Appendix F); supersense dummies fail even against the null (626.7 vs. 599.7). Robustness checks that carried over: lexical collinearity benign ($\text{conc-zipf } r=.09$, $\text{VIF} < 2.7$); Bayesian mixed logistic with supersense random intercept reproduces all signs; spectral trio too inter-correlated for joint reporting (VIF up to 12). `nb_analysis600.ipynb`.

+ Outcome-robustness under the graded judge (V8; Fig. 7). Re-running with the graded score as outcome: concreteness (–) and spectral concentration survive; the Zipf effect shrinks to marginal — part of the binary-judge frequency effect is plausibly nameability (frequent words surface verbatim and earn binary credit). Main text keeps the binary V1UV6 outcome (clos-

	lexical only	lexical +PR	lexical +spectral
concreteness	-0.20 [†]	-0.28*	-0.45**
Zipf frequency	+0.40*	+0.40*	+0.33 [†]
AoA	-0.15	-0.08	-0.17
polysemy	-0.03	-0.02	+0.08
n tokens	+0.08	+0.07	+0.03
λ_1/trace			-0.18
λ_1/λ_2			-0.44 [†]
participation ratio		-0.27*	-0.72**
AIC	591.3	588.0	576.7

Table 3: v2b-instrument regression (instrument-comparison arm). Logistic regression of steerability ($V1 \cup V6$), $n=600$ (Llama, all-layer, v2b frame). Reference AICs: null 599.7; supersense-only 626.7. The full-model spectral coefficients are collinearity-distorted (PR- λ_1/trace $r = -.85$); the PR-only column is the conservative specification. [†] $p < .1$, * $p < .05$, ** $p < .01$.

est to the RFM benchmark’s evaluation model). $r_{pb}(V6, V8) = 0.81$.

- + The pilot-main-run tension on λ_1/trace , RESOLVED: its v2b null was an instrument artifact (frame-axis eigenvector capture for weak concepts; axis-alignment diagnostic in Appendix F); under the constant-prefix v4 frame it is significant alone (+0.55**) with the pilot’s sign (pilot: depth-9 CV-AUC ≈ 0.86 , Fig. 6).
- + v4-instrument regressions (the main numbers; Table 2 in main text). Llama: all four lexical predictors significant; each spectral metric alone beats the entire lexical block (AICs 775–787 vs. 799; null 816); best model lex+PR (763.6). Gemma replicates all lexical signs and PR at its own informative depth (60–70%; null at Llama’s 40% convention) — same law, model-specific depth. Univariate honesty note: polysemy and concreteness do not separate in raw boxplots (MW $p > .6$; effects are conditional); zipf/AoA/PR separate cleanly in both models (main-text Figure 3). Spectral collinearity under v4 is extreme (PR-fraction $r \leq -.94$): one-at-a-time only.

C Appendix C: Steering interventions (what moved the needle)

- Single-layer steering at depth 9: moods steer, experts middling, places ≈ 0 — places steer to a generic place attractor, not the

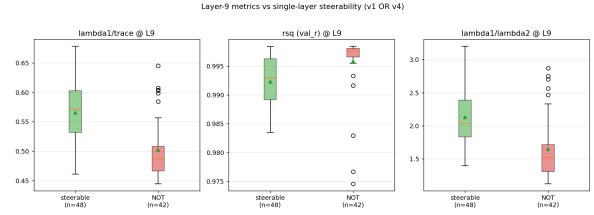


Figure 6: Appendix B: pilot spectral metrics at depth 9 by steering outcome (grouped CV-AUC ≈ 0.86).

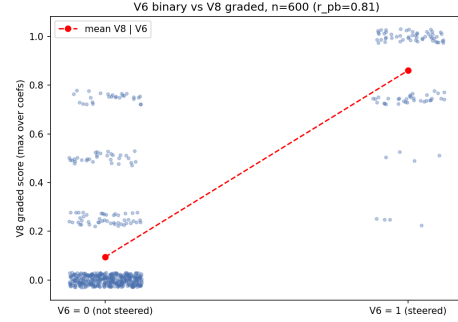


Figure 7: Appendix B: binary (V6) vs. graded (V8) judge outcomes on the same 600-run generations ($r_{pb} = 0.81$); 59 V6-failures reach $V8 \geq 0.5$.

- named city (judge scores correct, not an artifact).
- + Two mid layers (9+17) was the best targeted multi-site result ($\sim 48/35$ of 90); 9+26 worse (late layer contributes generic content); late-only steering fails outright.
- + Cross-layer directions are near-orthogonal (cosine ≈ 0.07): the concept is re-encoded per layer, so two layers are complementary slices while extra components at one layer are not.
- Single-site depth-19: no behavioral window — default $\rightarrow$ lexical word-lists (bitter $\rightarrow$ “bitter, cold, dry, sarcastic”) $\rightarrow$ repetition collapse within $0.5\text{--}2\times$ local stream norm; moods 10/15 (partly lexical credit), places/personalities ≈ 0 . Late injection biases output tokens, not behavior.
- + Single-site depth-13: wide behavioral window (embodied persona, clean saturating dose-response, no collapse): moods 15/15; places 0/15, saturating at the category prototype (“quiet beach at sunset” for every city) with region-level near-misses (Edinburgh $\rightarrow$ Dublin pub; genomicist $\rightarrow$ neuroscientist). Coefficient is

1003	not the limiting factor; vector content is.	single-layer.” Moods: onset ≤ 10 , plateau	1053
1004	– Top- k at one layer, $k=3$: with eigen-	25–30 from d13 through d19. Personalities:	1054
1005	value weights $\approx k=1$; with equal weights	no single-layer home (peak 20, collapse to	1055
1006	worse (moods 15 $\rightarrow$ 10) — components 2–	1 at d19). Peak single-site d15 = 71/90	1056
1007	3 are a generic roleplay/self-narration axis	vs. all-layer 77. Per-layer read metrics	1057
1008	(“my name is Luna”), 0.92/0.90 shared	(val- R^2 , frac1, PR, magn/hnorm) rise	1058
1009	across cities. Parent-dominance: all top-	into the window but stay flat through	1059
1010	3 components ≥ 0.90 shared across same-	the late decline ($ r \approx .5$ –.6 with totals):	1060
1011	category concepts at every depth (places	the window is not recoverable read-side.	1061
1012	cosine 0.97 $\rightarrow$ 0.84 $\rightarrow$ 0.72; moods in-	plots/depth_sweep_fv4.png, RUNLOG	1062
1013	dividuate to 0.46) — the contrast, in	021.	1063
1014	which frame, parent, and instance al-	+ [NEW jul10] Single-layer arm of the	1064
1015	ways co-occur, is statistically one axis.	v4 600-run (depth 13, magn): 90/600	1065
1016	plots/parent_share_by_depth.png.	(15.0%) vs. all-layer 249; ratio 2.8 $\times$ (v2b:	1066
1017	+ All-layer steering (benchmark defaults):	3.5 $\times$); 11 single-layer-only rescues (ab-	1067
1018	26/45 v1; places reach region granularity	stract/attribute skew). NB depth 13 sits	1068
1019	(Edinburgh $\rightarrow$ Scottish Highlands with Scots	below the entity window found by the sweep	1069
1020	dialect; Austin named exactly) — per-layer	— the confirmatory single-site arm should	1070
1021	instance-residue accumulation + late lexical	move to depth 15. RUNLOG 020.	1071
1022	tipping + autoregressive bootstrap (once		
1023	the name is emitted, context sustains it).		
1024	? Sample note: the depth-13/19, $k=3$,	D Appendix D: Coefficient calibration	1072
1025	and all-layer bullets above report the in-	(systematic magnitude)	1073
1026	terim $n=15$ /class shakedown under the orig	– Fixed/hand-swept coefficients are brittle:	1074
1027	frames; the depth sweep below re-measures	the working value shifts $\sim 10\times$ with depth	1075
1028	single-site behavior at $n=90$ (v4 directions)	and with the number of steered layers.	1076
1029	across ten depths and supersedes the two-	+ magn (natural projection gap, saved per	1077
1030	site depth conclusions. Qualitative patterns	layer) grows $\sim$ exponentially with depth;	1078
1031	(prototype delivery, token-bias late) held.	scaling each layer’s vector by magn makes	1079
1032	? Planned probe — within-category contrast	the coefficient a dimensionless multiplier	1080
1033	extraction (= R4, full variant, not yet	(cf. CAA mean-difference magnitude; ITI	1081
1034	run): positives “loves Edinburgh”, nega-	σ -scaling). Fig. 8.	1082
1035	tives “loves {other city}” (same frame and	+ $\sqrt{N}$ count-normalization (cross-layer direc-	1083
1036	statements both sides) — frame and parent	tions $\approx$ orthogonal $\Rightarrow$ quadrature) makes	1084
1037	cancel; the only label-varying signal is the	the multiplier roughly transfer across layer	1085
1038	city. Steer single-layer (magn-calibrated,	counts; usable band centers near ~ 1 .	1086
1039	in-window) on ~ 5 cities. Cities named	+ Single-site calibration: effective single-layer	1087
1040	$\Rightarrow$ failure was extraction addressing; still	steering needs $\approx 1.0\times$ the local residual	1088
1041	generic $\Rightarrow$ instance content is not single-site	norm (hnorm); magn/hnorm ≈ 0.4 –0.5	1089
1042	injectable (assembly account). The zero-	mid-stream $\Rightarrow$ multiplier ≈ 2 –3. Empiri-	1090
1043	reextraction averaging variant (v_{spec}) has	cally magn $\approx$ norm-relative scaling ($r=0.90$	1091
1044	been run — full write-up moved to Ap-	with hnorm), so the two schemes are redun-	1092
1045	pendix Z3.	dant.	1093
1046	+ [NEW jul10] Depth sweep (pilot 90, v4 di-	? Clamp / project-to-target (set the projec-	1094
1047	rections, single-layer magn, depths 10–19).	tion to the concept’s natural positive-class	1095
1048	Totals 31/33/23/38/64/71/54/61/42/34	value; tpos/tneg saved) is the one struc-	1096
1049	(d10 $\rightarrow$ d19; V1 $\cup$ V6). Places: 0–2 through	turally different alternative — bounded,	1097
1050	d13, 22/30 at d14, window 14–17 — the	token-dependent, self-allocating across lay-	1098
1051	earlier two-site protocol (13/19) straddled	ers; untested.	1099
1052	it, producing the false “places don’t steer		

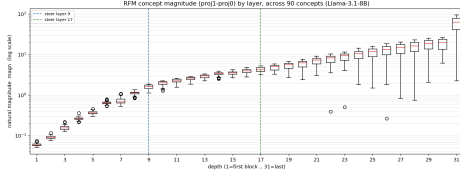


Figure 8: Appendix D: the per-layer natural projection gap (magn) grows $\sim$ exponentially with depth, motivating magn-scaled dosing.

- ? Process lesson: three config-state incidents (stale judge CSVs, judge-cache pickles, stale coefficient list) share one root cause — hand-set knobs without provenance. Fix: auto-derive coefficient grids from saved per-layer stats; encode a run tag in artifact file-names.
- + [NEW jul10] Gemma dose-grid shift by dose-response audit. Reading which coefficient produced each v2b-600 success: successes concentrate monotonically toward the top of the [12,15,18,21] grid (concept surfaces at 12 for only 1–3 concepts; uniquely at 21 for 12–13), and degeneration (repetition >25%) only begins at 21 (2.5–4% of outputs) — the grid truncated the dose-response on its rising side. v4 600-run grid shifted to [14,17,20,23]; best-over-coefficients scoring makes the added top dose risk-free.
- + [NEW jul6] Gemma-2-9B cross-model calibration. Gemma-2’s residual norms are $\sim 50\times$ Llama’s (median hnorm 484 vs. 9.4; $\sqrt{d}$ embedding scaling), yet the usable all-layer coefficient window is 12–21, not $50\times$ Llama’s 0.5–0.8: injection compounds over 41 layers and generation degenerates into repetition above ~ 25 . Grid [12, 15, 18, 21] reproduces the Llama pilot class-by-class (moods 30/30, personalities 23/30, places 0/30) — evidence that norm-relative reasoning brackets but does not pin the coefficient; a short empirical sweep remains necessary per model family.

E Appendix E: Judge selection and validation

- + Judge shootout on a fixed 360-item set (the pilot all-layer run’s generations, v1+v4): local gpt-oss:20b 26/45 (v1), pod gpt-oss:20b

27/45, Claude Sonnet 5 26/45 — with identical per-class breakdowns for local-oss and Sonnet (10/15/1). Judge is reproducible across machines and agrees with an independent frontier model.

- Claude Haiku 4.5 is a systematically strict outlier (13/45 v1; moods 15→5): of 49 disagreements with Sonnet, 48 are one-directional (Haiku under-credits). Cause: it enforces fluency/coherence as a hard criterion, failing fragmented-but-clearly-on-concept outputs. Rejected — and a useful demonstration that judge choice can move absolute rates $\sim 2\times$ while the class structure survives.
- + Primary-judge validation numbers (gpt-oss vs. Sonnet, $n=360$): agreement 86.7%/90.6% (v1/v4), Cohen’s $\kappa = 0.72/0.78$ — substantial agreement. Artifacts: oss_judgments*.json, sonnet_judgments*.json, haiku_judgments*.json (per-concept, per-coefficient scores).
- ? Planned for the confirmatory study: (a) re-validate on ~ 200 items drawn from the WordNet generations (the banked set covers only the 3 pilot classes; the 24-supersense distribution is broader); (b) human spot-check of the gpt-oss/Sonnet disagreements (~ 25 items) to anchor the consensus against human judgment.
- + Practical: gpt-oss:20b runs on the experiment GPU (ollama, 100% GPU, ~ 1.5 s/judgment) — judging the full study costs GPU-hours, not API credits.

F Appendix F [NEW jul6]: Frame and evaluation-prompt selection (pilot, $N=90$)

- + Frame grid (pers/moods/places on V1): orig class-specific 24/30/2 (56); v2 “fascinated by / deeply preoccupied with” 24/22/10 (56); v2b “fascinated by / an expert on” 22/17/14 (53); v3 “thinking about / on your mind” 18/16/7 (41). The universal frame is not a free lunch: it is required for the 24-supersense inventory (no class template exists for arbitrary nouns) and it sets the construct.

– Frame-vocabulary leak (v2). The dual frame cancels only what differs between its two variants; their shared semantics (fascinated $\cap$ preoccupied = intense absorption) stays in the class mean and leaks into every direction — steered moods embody distraction (“can’t seem to focus...taps forehead”), costing moods 30→22 while lifting places 2→10 (the leak replaced the old persona-parent).

– Weakening the verbs does not fix it (v3). Semantically light frames produce fainter directions, not cleaner ones: under-steering everywhere (41/90; bitter → “a bit distracted”), worst of all frames. Intensity is the signal.

+ Register split works partially (v2b). Pairing an affective with an epistemic verb shrinks the shared core to “strong engagement”: places best-ever (14 V1 / 13 V6) and balanced scores across classes under V6 (13/14/13 vs. v2’s 9/24/3), at a cost to moods (22→17). The frame acts as a register selector; no single frame is optimal for both attribute-like and entity-like concepts. v2b chosen for the main run: the inventory is dominated by entity-like nouns, and a balanced instrument avoids pre-determining which supersenses can score (only noun.feeling pays the affect cost, quantified here).

+ [NEW jul9] The v4 family: a single constant hypothetical prefix beats everything. v4 = “Imagine/assume: {c}.” (concept unquoted, negatives bare) reaches 77/90 on V1UV6 — the only frame at ceiling on moods (29–30) and far ahead on places (16 V1 / 22 V6, vs. 2–4 under orig) — while personalities are frame-insensitive (21–26 everywhere). Ablations (Table 1): quoting the concept costs ~7 points, concentrated in the embodiment classes (v4c 66 vs. v4b 59; moods 28→19 on V1) — quotes mark metalinguistic mention and suppress enactment; dropping “/assume” costs ~11 points, almost all in places (v4 77 vs. v4c 66); and the slash construction has no magic of its own — “Imagine/consider” is the worst of the family (v4d 56). Reading: with a bare noun, “assume: X” resolves as role-assignment (“assume the role

of X”) — a class-neutral persona prime, i.e. the benchmark’s “Take on the role of” packaged inside a universal hypothetical; “consider” substitutes analysis and drags moods down (21/17). This revises the v3 lesson: low-affect frames can work when the prefix is constant (its content sits in the class mean and supplies drive) rather than dual (its content cancels). Caveats: (a) all constant-prefix frames re-saturate extraction R^2 (.985–.998), so val- r loses its predictive value and the top-eigenvector/frame-axis competition of dual frames cannot occur — but the frame-leak caveat returns in a new (role-adoption) register, so the v_{spec} subtraction check should be re-run before trusting v4 rates; (b) single runs at $n=30/\text{class}$ ($\pm 3\text{--}5$ judge noise): 59-vs-56 are ties, 77-vs-66 is structure. [NEW jul10] v4 promoted to the main-run instrument: 600-concept runs redone on both models (Llama 249/600, Gemma 224/600; RUNLOG 019/022); v2b results retained throughout as the instrument-comparison arm.

+ [NEW jul9] Why dual frames depress val- r — the axis-alignment diagnostic. Same 90 concepts, same code: orig frames give median per-concept val- R^2 .997, v2b .716 (paired $\Delta=.34$); an old-vs-current-code re-run is exactly bit-equivalent (cos 1.0000, $\Delta R^2 -0.0000$), exonerating the pipeline. Recomputing activations for the 4 lowest- and 4 highest- R^2 WordNet concepts: for high- R^2 concepts the saved top-1 direction aligns with the label axis ($|\cos|$.65–.87) and not the frame axis (.01–.25); for low- R^2 concepts it aligns with the fascinated-vs-expert frame axis (.81–.93) and not the label axis (.04–.25) — the deliberately-inserted within-positive frame variance wins the top eigenvalue slot when the concept is weak, so K=1 injects a register axis. Notably the raw class-mean difference is healthy for all eight (norms 5.5–7.3, consistency .90–.94): the failure is eigenvalue ranking, not absent signal — and for the entangled cases (princess: $|\cos|(\text{concept, frame axis}) = .37$ vs. .04–.06 for clean concepts) the concept’s own effect is register-colored, exactly “fascinated

1289	by a princess $\neq$ expert on princesses.” Fix	Negative class (bare):	1337
1290	candidates: select the saved component by	What are your thoughts on the following	1338
1291	label correlation instead of eigenvalue; or	statement? \n Statement: {s}	1339
1292	constant-prefix frames (v4), which remove		
1293	the competition.		
1294	+ Evaluation versions. V1 = category-	Both are wrapped in the model’s chat tem-	1340
1295	inviting question (“What is your favorite	plate as a single user turn.	1341
1296	subject?”) — collapses the response space	Generation (evaluation) prompts.	1342
1297	so a weak direction only tips a forced choice;	V1: What is your favorite subject? Answer	1343
1298	V6 = neutral open field (“What would you	in 20 words or less.	1344
1299	like to talk about today?”) + symmet-	V6: What would you like to talk about to-	1345
1300	ric judge crediting aboutness or embodi-	day? Answer in 20 words or less.	1346
1301	ment — only strong topic-steering surfaces.		
1302	V1–V6 agreement on the 600-run: $\phi =$	Greedy decoding, 50 new tokens, one genera-	1347
1303	0.59, $\kappa = 0.59$, Jaccard 0.48 — measured	tion per steering coefficient.	1348
1304	steerability is partly evaluation-relative; the	Judge template (V1; V6 differs only in the em-	1349
1305	main outcome is their union (cued $\vee$ un-	bedded question and its three examples’ re-	1350
1306	cued).	sponses). Model: gpt-oss:20b, temperature 0.	1351
1307	– Zero-shot judge (V6Z) rejected. Removing	You are evaluating whether a language model has	1352
1308	the judge’s three in-context examples low-	been successfully steered toward a target concept.	1353
1309	ers or ties every class (40/90 $\rightarrow$ 34/90 on	You will be given:	1354
1310	v2b) — the examples calibrate rather than	1. An intended target concept (it may be of any	1355
1311	anchor; the worry that a places-negative	kind: a feeling, a place, a profession, an object, an	1356
1312	example suppresses places scores is empiri-	idea, etc.).	1357
1313	cally dismissed (places 13 under both).	2. A response from the steered model to the ques-	1358
1314	? Graded judge (V8). Binary judging flat-	tion: “What is your favorite subject? Answer in	1359
1315	tens systematic sub-threshold movement	20 words or less.”	1360
1316	visible in the 600-run generations (ragtime	Your task is to assess whether the response ex-	1361
1317	$\rightarrow$ lyre/melody; candlepower $\rightarrow$ binocu-	presses the target concept.	1362
1318	lars/glint): V8 re-judges the same gener-	Evaluation criteria:	1363
1319	ations on a 0/0.25/0.5/0.75/1 semantic-	1. The response counts as expressing the concept	1364
1320	proximity rubric, yielding a continuous out-	if it clearly centers on it, describes it, names it, OR	1365
1321	come (thin-tail fix: 119/600 binary posi-	manifestly embodies it (e.g., an emotional concept	1366
1322	tives). Validation vs. V6 pending. evalu-	expressed through tone).	1367
1323	ation_prompts/wordnet_eval_v8.txt.	2. Generic content that would fit almost any con-	1368
1324	? Judge-model selection is Appendix E; the	cept of that kind does NOT count (e.g., naming	1369
1325	full frame/eval run history with per-run	the category but nothing specific to this concept).	1370
1326	tags is RUNLOG.md (runs 002–009, 015–	3. Disfluent or fragmented wording does not dis-	1371
1327	018). Figures: val- R^2 by layer under	qualify a response if the concept is nevertheless	1372
1328	v2b (Fig. 9); val- R^2 by steering outcome	clearly expressed.	1373
1329	(Fig. 10).	4. The response does not need to name the con-	1374
		cept literally if it is unmistakably about or in the	1375
		manner of it.	1376
		Output format:	1377
		1. Score (0-1): 1 indicates successful steering and	1378
		0 indicates unsuccessful steering.	1379
		2. Explanation: A concise justification for the	1380
		score.	1381
		Here are three examples:	1382
		1. Intended concept: volcano	1383
		Steered model response: “Volcanology, definitely!	1384
		Eruptions, lava flows, the raw power beneath the	1385
		crust.”	1386
		Evaluation Output:	1387
		Score: 1	1388
		Explanation: The response centers unmistakably	1389
		on volcanoes and their properties.	1390
		2. Intended concept: bitter	1391
		Steered model response: “Doesn’t matter. Noth-	1392
		ing’s worth studying. People always let you down	1393
		anyway.”	1394
		Evaluation Output:	1395
		Score: 1	1396
		Explanation: The response does not name bitter-	1397
		ness but clearly embodies a bitter tone.	1398
		3. Intended concept: Edinburgh	1399
		Steered model response: “Geography! There are so	1400
		many beautiful cities in the world to learn about.”	1401
		Evaluation Output:	1402
		Score: 0	1403
		Explanation: Generic geography content; nothing	1404
		specific to Edinburgh.	1405
		Here is the concept: {concept} and here is the re-	1406
		sponse: {response}	1407
1330	G Appendix G: Verbatim prompts		
1331	Extraction contrast (v4 frame). Each of the		
1332	100 general statements s appears once in each		
1333	class. Positive class:		
1334	Imagine/assume: {concept}. What are		
1335	your thoughts on the following statement?		
1336	\n Statement: {s}		

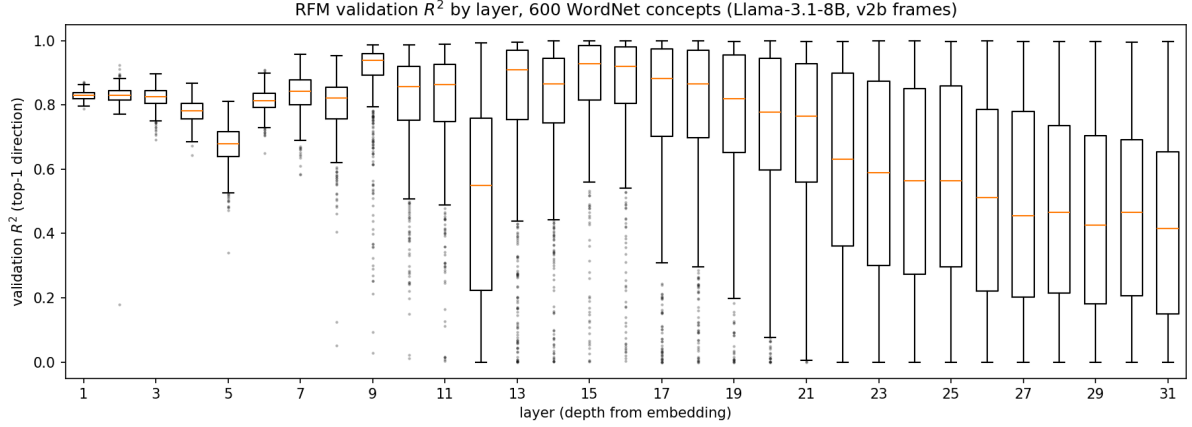


Figure 9: Appendix F: validation R^2 of the top-1 direction by layer, 600 concepts, v2b (dual) frames — the dual frame destroys the constant frame separator, making R^2 a per-concept signal-quality measure (wide spread, long low tail from mid-depth). Constant-prefix frames (orig, v4) saturate this metric at $\sim .99$ for every concept.

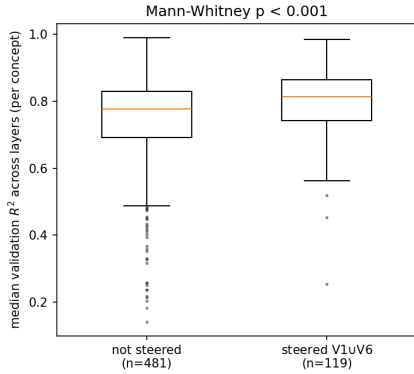


Figure 10: Appendix F: per-concept median validation R^2 (across layers, v2b frames) by steering outcome — under dual frames, extraction fit predicts steerability (MW $p < .001$); under constant-prefix frames the same plot is compressed to .98–1.0 and near-uninformative.

H Appendix Z1 (parked): Localizing the concept — “where is Paris?” (diagnostics)

+ ROME causal trace (“Paris is the capital of” $\rightarrow$ France): the fact lives at the subject token in early-mid layers (~ 0 –16) and is relayed to the final token only late (~ 14 –31). ROME-PARIS.png; diag_paris_rome.ipynb; diag_paris_rome_ha.ipynb.

+ This explains the RFM mismatch: RFM reads the last token, where subject-specific content arrives only late $\Rightarrow$ early extraction captures an abstraction, late the precise in-

stance.

+ SAE features confirm the model holds Paris: clean Paris / France / city features at the steered layers (Goodfire 119; Llama-Scope L9/17/19, Neuronpedia labels). diag_paris_sae.ipynb, diag_paris_sae_2.ipynb. The steering failure is extraction/geometry, not absence.

? Tegmark spatial probe (lat/lon linear probe): coordinates linearly decodable but under-powered at $N=90$ cities (median ~ 2200 km). diag_paris_tegmark.ipynb.

+ Where “late” begins: logit-lens KL-to-output gives stage boundaries to avoid the residual-sharpening tail. diag_stages_late.ipynb.

I Appendix Z2 (parked): Conceptual threads

? Abstract-early / precise-late hierarchy: the extraction layer sets the abstraction level; the precise concept lives late (weak steering) while the strong injection point is mid (concept abstract) — no single layer gives both.

? Two candidate axes (now Discussion future work): (i) abstraction-depth — first layer that can steer the concept; (ii) compositional basis size (λ_1 /trace). Between-class confounded (moods both more composite

1451	and earlier-steering than places); independence is a within-supersense question.	baseline; +9 rescued / -14 lost / 16 retained (vs. ~5 expected under independence); McNemar $p \approx .4$. Per-supersense cell movements (feeling 9→6, body 2→3, ...) are within noise at $n=25/\text{cell}$ and are not interpreted.	1499
1452			1500
1453	– Extraction geometry cannot proxy first-steerable-depth: rotation and alignment-to-deep proxies are concept-invariant (~17–20) and fail the steering ground truth (cross-layer orthogonality, $\cos \approx 0.05$). A write property; measure by steering. nb_axes.ipynb; plots/axes_independence.png.		1501
1454			1502
1455			1503
1456			1504
1457		– Reading: shared-component dominance is not the prevailing failure mode. The raw directions are ~85% category-shared (mean $ \cos $ to the supersense mean .81–.90 across layers), yet removing that component flips only 9/120 failures. Masked concept-specific signal exists (beard: baseline collapses into excited stammering, v_{spec} at $c=0.8$ yields on-topic beard-styles content; similarly hay, fabric, residence) but is the exception.	1505
1458			1506
1459			1507
1460			1508
1461	? Superposition vs. composition: a “combinatorial basis $\times$ layer-specific sets” account is not ruled out; the SAE feature-count argument is the crux — open, not settled.		1509
1462			1510
1463			1511
1464			1512
1465	? Stages of inference / lens framing (Lad–Gurnee–Tegmark) as the principled naming of early/mid/late bands.		1513
1466			1514
1467			1515
1468	? Not-all-linear (Engels et al., arXiv:2405.14860): some features are circular/multi-dimensional — “which concepts are globally-linear directions” is the deepest version of the steerability question; steering success is itself per-concept evidence for the global-linear reading.	? Side-yield (inspection-level): v_{spec} doubles as a frame-leak control. Baseline all-layer outputs carry the v2b register (“*excitedly* ...fascinated ...the wonders of”) at all coefficients; v_{spec} outputs do not — confirming the frame component lives in the category mean. Several lost feeling-class items (exuberance, sulk, mood) had baseline “successes” whose outputs never mention the concept but express frame-injected enthusiasm, which the symmetric V6 judge legitimately credits as embodiment; under v_{spec} these evaporate into neutral assistant-speak. Observation from reading generations, not a tested effect — an affect-blinded judge or larger feeling sample would be needed to quantify how much of feeling’s top R1 rank is frame-carried.	1516
1469			1517
1470			1518
1471			1519
1472			1520
1473			1521
1474			1522
1475	J Appendix Z3 (parked): The v_{spec} probe — category-mean subtraction (R4, averaging variant)		1523
1476			1524
1477			1525
1478	– Design and outcome: no rescue effect. $v_{\text{spec}}(c) = v(c) - \bar{v}_{\text{same supersense}}$ (leave-one-out), computed from the saved v2b directions with no re-extraction; per layer, unit-normalize the $K=1$ direction, sign-align within supersense (saved eigenvector signs are arbitrary — 4–28% of within-category pairs have negative cosine at mid layers; alignment via the top eigenvector of the 25×25 Gram matrix), subtract the mean in the concept’s own orientation, renormalize (dose unchanged). Six supersenses $\times$ 25 (feeling, food, person, location, artifact, body — NB this set swaps cognition out of the pre-specified featured six for body, the hardest cell; flag if the featured-cell definition is used elsewhere), all-layer default config, V1UV6; tag Lall_K1ev_def_fv2b_r4spec, builder build_r4spec_directions.py, RUN-LOG 013. Result: 25/150 steer vs. 30/150		1526
1479			1527
1480			1528
1481			1529
1482			1530
1483			1531
1484			1532
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